

Market Data Research and Trading Strategy Evaluation

Optiver order-book research and an ETF trend-filter backtest
Research memorandum | Florian Siakoua Toukam | October 7, 2026

Investment research question

Can order-book and trade information improve a short-horizon volatility forecast, and can a simple trading rule reduce market risk without giving up too much return? I examined these questions separately because the available datasets support different tests.

What the analysis shows

Research test	Result	Implication
Five-minute volatility	Ridge test RMSPE 28.67% versus 34.22% for persistence.	The model improved the basic forecast. Stronger baselines substantially narrow the gain.
SPY trend filter, 2020-2026	Net CAGR 10.36% versus 15.29% for buy-and-hold.	The rule reduced risk but surrendered return in the later historical period.
SPY downside	Maximum drawdown -23.20% versus -33.72%.	Risk reduction is meaningful, although a 50% stock/cash benchmark had a smaller drawdown.

Conclusion

I would treat the volatility model as a risk-estimation input for further research. Its advantage over a calibrated simple forecast is too uncertain to describe as a durable edge. I would treat the trend filter as an exposure-management rule whose return cost must be justified by the investor's risk objective.

The strongest finding is the distinction between reducing risk and improving investment returns. Neither a better volatility forecast nor a lower drawdown, by itself, establishes a profitable trading strategy.

Scope: a newly reconstructed academic research case using public historical data. The simulations are not professional trading, actual executions, or work performed for Optiver or SG. ETF test period ends September 30, 2026; 2026 is partial.

1. Order-book and trade research

I processed 3,694,546 quote records and 698,943 trade records for anonymized stock IDs 0, 1 and 3. The six parquet files are saved with source URLs, a fixed mirror commit, and checksums. Stock identities and calendar dates are not supplied.

For each observed ten-minute bucket, I used seconds 0-299 to build predictors and seconds 300-599 to measure the target. The target is second-half realized volatility based on top-of-book weighted-average-price log changes. This is a custom five-minute task, not the competition's original subsequent-ten-minute target.

Predictors cover prior realized volatility, relative bid-ask spread, displayed depth, order-book imbalance, quote-observation count, trade volume, trade-record count and execution count. Numeric inputs are transformed where appropriate; Ridge scaling is fitted on training data only.

The eligible sample contains 11,488 stock-bucket observations. Two buckets failed the coverage/positive-volatility criteria. I allocated complete time-ID groups to training (8,041 rows), validation (1,722) and test (1,725). All stocks from a simultaneous bucket remain together.

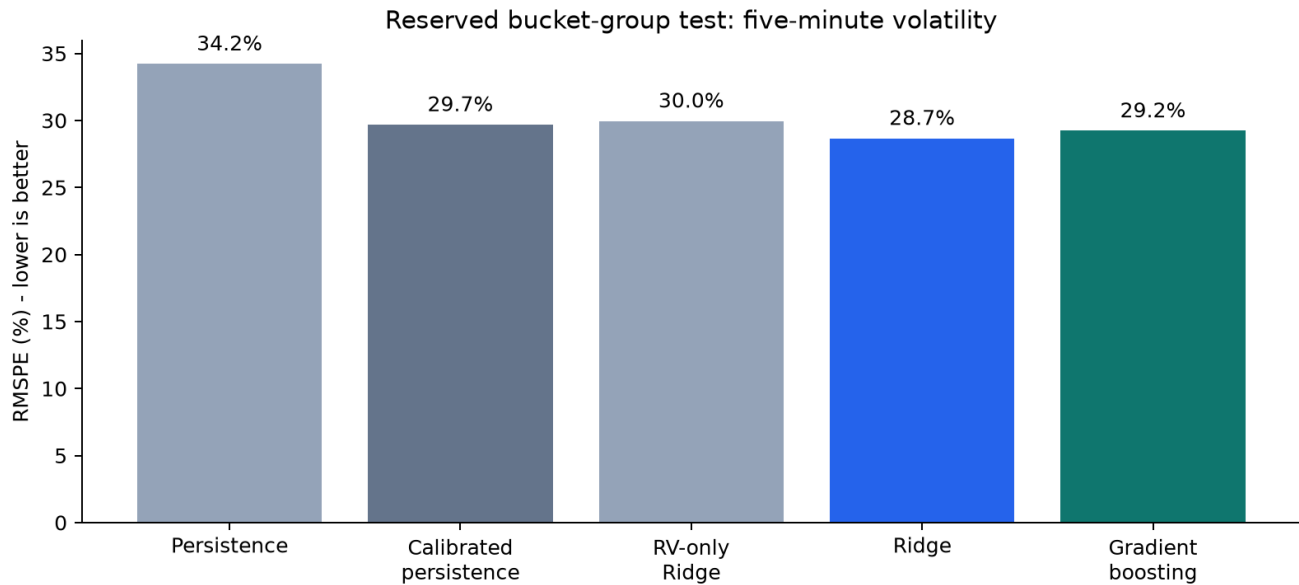
The bucket IDs are shuffled. This is a reserved group test, not a chronological out-of-time test. It prevents the same simultaneous bucket from appearing in both training and test data, but cannot establish future-calendar generalization.

Forecast comparison

Forecast	Validation RMSPE	Test RMSPE
Persistence	32.86%	34.22%
Calibrated persistence	28.69%	29.69%
RV-only Ridge	28.29%	29.96%
Ridge	26.35%	28.67%
Gradient boosting	26.80%	29.23%

Persistence repeats the first-half volatility. Calibrated persistence multiplies that forecast by a training-only factor chosen to minimize relative squared error. RV-only Ridge uses previous volatility and stock identity; the full Ridge adds market-data features. The two full models have fixed settings, and validation RMSPE selects Ridge over gradient boosting.

2. How much value did the added data contribute?



The full Ridge model reduced test RMSPE by 16.2% relative to uncalibrated persistence. Against calibrated persistence, the improvement was 3.4%. Against RV-only Ridge, it was 4.3%.

The 95% bucket-group bootstrap interval for improvement versus calibrated persistence runs from -2.6% to 8.6%. It includes no improvement. The corresponding interval versus RV-only Ridge is 0.9% to 7.1%.

That changes the interpretation. Much of the improvement over the naive forecast can be obtained by calibration. The additional market-data features contribute a smaller gain in this sample. I would prioritize the simple model and stronger temporal validation before adding complexity.

What the forecast can support

A short-horizon volatility estimate can inform how a trader evaluates changing risk, position size, or execution conditions. This study does not test those decisions directly. It forecasts price variability, not price direction, fill quality, or profit.

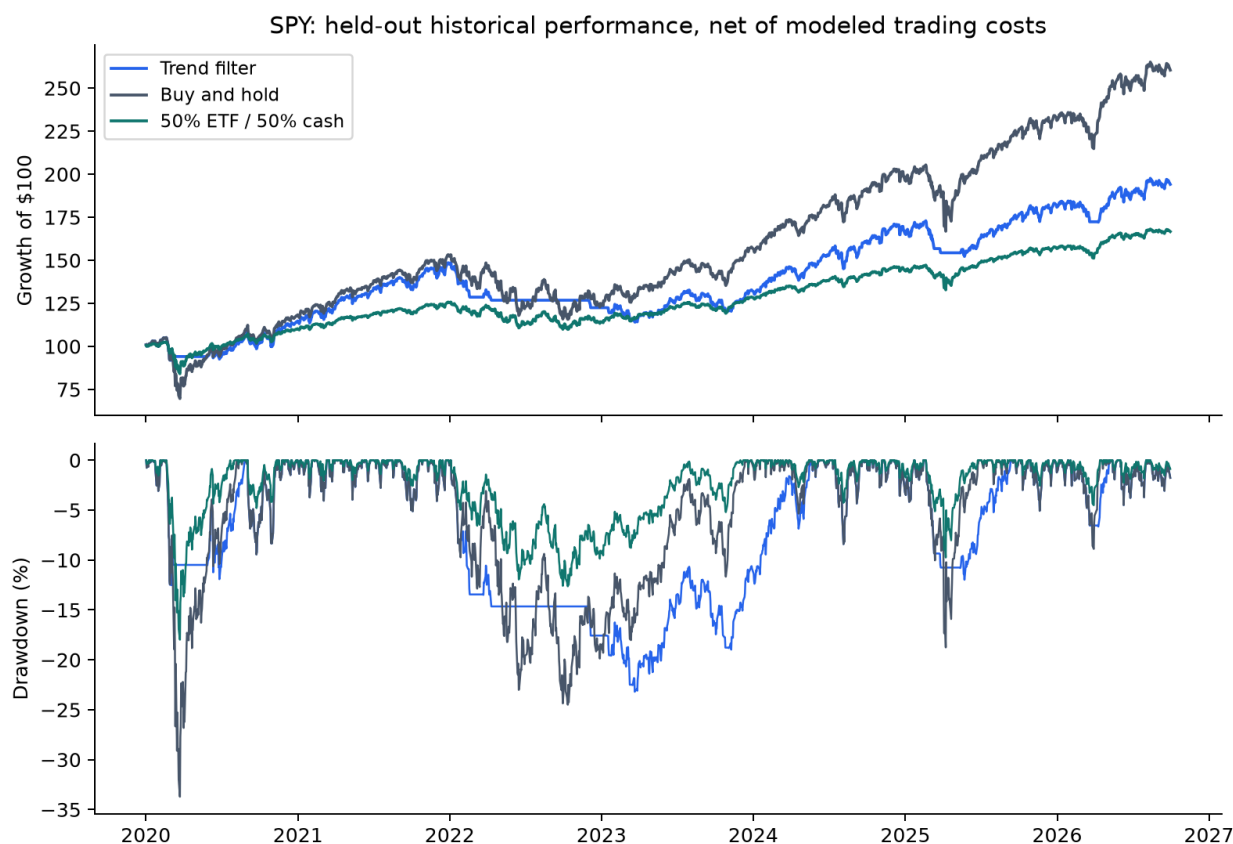
The bootstrap keeps simultaneous stocks together but treats the anonymized buckets as exchangeable. Unknown calendar ordering prevents a serial-dependence-aware temporal audit. Three anonymized stocks also limit how broadly the result can be applied.

3. Trading-strategy test

I tested a fixed, long-only SPY rule. A daily close above its trailing 200-session price average generates a long signal; a close at or below the average generates a cash signal. Each decision executes at the following session's close. The new position first earns the next close-to-close return.

The model charges 5 basis points on one-way traded notional, has no leverage, and assumes cash earns zero. These are explicit research assumptions. Returns use Yahoo adjusted closing prices as a dividend/split-adjusted total-return proxy. A separate daily-rebalanced 50% ETF/50% cash benchmark helps distinguish timing from simply holding less market exposure.

Daily price history begins January 2, 2001. The simulation begins January 2, 2002 after the rolling-average warm-up. Development is 2002-2014, validation is 2015-2019, and the later historical evaluation is January 2, 2020 through September 30, 2026. The 200-session rule was fixed before reviewing results.



Later historical period	Net CAGR	Volatility	Max. drawdown
Trend filter	10.36%	12.41%	-23.20%
Buy and hold	15.29%	20.05%	-33.72%
50% ETF / 50% cash	7.89%	10.02%	-17.98%

CAGR and volatility use 252 trading sessions per year. Drawdown includes the period's starting value. Positions carry across research-period boundaries; all portfolios liquidate on the final date. All transactions shown are simulated.

4. Robustness and decision implications

The primary rule was invested for 79.0% of the later period. It reduced SPY's annualized volatility from 20.05% to 12.41%, but its CAGR was 4.93 percentage points lower. The simple 50% stock/cash portfolio had lower volatility and a smaller drawdown, with lower return.

The historical regime matters. In 2002-2014, the trend filter produced a 6.86% CAGR versus 6.59% for buy-and-hold, with much lower maximum drawdown. In 2015-2019, its CAGR fell to 6.68% versus 11.58%. The stronger early result did not carry through consistently.

Transfer checks: the same rule and costs

ETF	Trend CAGR	Buy/hold CAGR	Trend drawdown	Buy/hold drawdown
SPY	10.36%	15.29%	-23.20%	-33.72%
QQQ	19.05%	21.06%	-21.92%	-35.12%
IWM	6.55%	9.32%	-21.06%	-41.07%

The same trade-off appears across all three ETFs: lower observed drawdown, but lower CAGR than buy-and-hold. This supports a risk-management interpretation rather than a broad claim of return outperformance.

Cost and rule sensitivity

In the SPY later-period test, increasing one-way cost from 0 to 25 basis points reduced the primary rule's CAGR from 10.72% to 8.96%. At 5 basis points, the 50-, 100-, 200- and 250-session rules produced CAGRs of 8.48%, 12.00%, 10.36% and 8.45%. These are robustness comparisons; the best result was not substituted for the primary rule.

An additional one-session execution delay produced 9.87% CAGR; an additional two-session delay produced 12.15%. A constant illustrative 3% cash yield produced 11.05%. The cash-yield test is not a reconstruction of historical Treasury or money-market returns.

A 20-session moving-block bootstrap interval for annualized mean return difference versus buy-and-hold includes both positive and negative values. It does not establish a statistically reliable return advantage. Historical knowledge may also influence the choice of research question; this holdout is not prospective live testing.

5. Research conclusion and verification

I would retain the order-book study as evidence of a complete research process: check the input data, define information available at the decision point, compare simple and more complex forecasts, reserve evaluation groups, and explain the result against a meaningful baseline.

I would retain the ETF test as evidence of investment judgment. A rule can make a portfolio less volatile and still leave the investor worse off on return. The decision depends on the purpose of the portfolio, tolerance for drawdown, the value of time in cash, and implementation costs.

The next useful research step would be dated intraday data, temporal model validation, and a specific execution or position-sizing decision tied to the volatility forecast. This project stops at the evidence currently supported by its datasets.

Checks completed

Source-file checksums and record counts agree. SQL checks found no duplicate feature keys, split leakage groups, invalid volatility targets, or duplicate ETF dates. Changing future quotes and trades did not change first-half features. Changing future ETF prices did not change earlier signals.

A separate cash-and-fractional-holdings ledger agrees with the vectorized strategy accounting, including transaction costs and both benchmarks. An execution-timing fixture confirms that a new position cannot earn the return before its modeled entry. CAGR and drawdown were independently recomputed from the saved daily returns.

Sources and reproducibility

Optiver data: [Kaggle competition dataset](#). Files retrieved from a [fixed-commit public mirror](#). The manifest records each original path and checksum. The mirror is a third-party copy, not a live Optiver data feed.

Bucket structure: [Optiver host dataset clarification](#). Time IDs are shuffled; the original target uses the following ten minutes.

ETF prices: Yahoo Finance chart API snapshots retrieved October 7, 2026. Raw JSON, corporate actions, and daily CSVs are retained. Dividend dates were checked; adjusted returns can differ slightly from a simple cash-dividend calculation because of adjustment conventions.

ETF context: [State Street SPY information](#). **Methods:** [scikit-learn validation guidance](#), [Ridge documentation](#), and [Faber's moving-average research](#) for background. This implementation uses its own stated daily timing and cost assumptions.

Reproduction: run the saved scripts against the cached snapshots. Downloading fresh prices can produce different adjusted-price histories. Source and model settings are recorded in `research_plan.json`; all principal outputs are in `04_Results`.